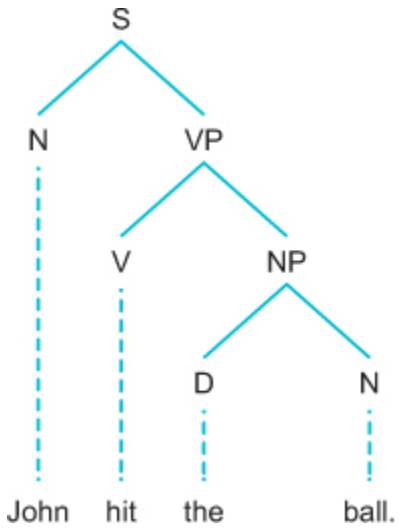
representation is a form of *compositionality*, meaning that a complex piece of data is represented as the combination of more elementary components, and those components can be further broken down into even simpler components, and so on, until you get to atomic units.

Human language is compositional (figure 1.3). For example, a book is composed of chapters, chapters are composed of paragraphs, paragraphs are composed of sentences, and so on, until you get to individual words, which are the smallest units of meaning. Yet each individual level conveys meaning—an entire book is meant to convey meaning, and its individual paragraphs are meant to convey smaller points. Deep neural networks can likewise learn a compositional representation of data—for example, they can represent an image as the composition of primitive contours and textures, which are composed into elementary shapes, and so on, until you get the complete, complex image. This ability to handle complexity with compositional representations is largely what makes deep learning so powerful.

Figure 1.3. A sentence like "John hit the ball" can be decomposed into simpler and simpler parts until we get the individual words. In this case, we can decompose the sentence (denoted S) into a subject noun (N) and a verb phrase (VP). The VP can be further decomposed into a verb, "hit," and a noun phrase (NP). The NP can then be decomposed into the individual words "the" and "ball."



1.2. Reinforcement learning